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Paleomagnetism and K-Ar Ages of Volcanic Rocks From Long Valley Caldera, California

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Paleomagnetic measurements and K-Ar age determinations on volcanic rocks from Long Valley caldera, California, have enabled further refinement of eruptive activity within this large silicic volcanic center. K-Ar age determinations show that postcaldera volcanic eruptions began 0.73 m.y. ago and continued periodically until about 50,000 years ago. The eruptions were not temporally random but tended to occur in distinct episodes separated by periods of quiescence. Volcanism in the western half of the caldera was particularly intense between 0.15 and 0.50 m.y. ago, when many units ranging in composition from basalt to rhyolite were erupted. An average of the virtual geomagnetic poles for 33 units from the caldera yields a paleomagnetic pole at 89.7°N, 138.4°E, (α_{95} = 5.1°), which is indistinguishable from the earth's rotational axis. The ancient geomagnetic field dispersion about this mean pole is 16.0°, with upper and lower limits of 19.3° and 13.6°, respectively.

INTRODUCTION

Long Valley, California, is a major topographic depression on the east side of the Sierra Nevada about 30 km south of Mono Lake. *Gilbert* [1938, 1941] identified Long Valley as the likely source of the voluminous Bishop Tuff and related ash deposits in the western United States and suggested that the depression itself was related to the extrusion of magma from beneath the valley. Geophysical studies by *Pakiser* [1961] and *Pakiser et al.* [1964] outlined the extent and depth of this depression. *Smith and Bailey* [1968] recognized Long Valley as a resurgent caldera. It is thought to be one of only three calderas in the western United States (along with Valles caldera in New Mexico and Yellowstone caldera in Wyoming) of sufficient size and youth to contain residual magma in their chambers [*Bailey et al.*, 1976].

This paper presents primarily the results of a paleomagnetic investigation of volcanic rocks within the caldera. The paleomagnetic data are supplemented by several new potassium-argon age determinations and a recalculation of age data previously published [*Bailey et al.*, 1976] using revised ^{40}K decay constants [*Steiger and Jäger*, 1977]. The paleomagnetic study was undertaken to (1) search for the Blake Reversed-Polarity Subchron or other reversed events within the Brunhes Normal-Polarity Chron; (2) refine the timing of postcaldera eruptions in Long Valley; (3) obtain information about paleosecular variation in the western United States; and (4) investigate the magnetic properties of various silicic rocks that have hitherto been little studied.

SUMMARY OF VOLCANIC HISTORY

Volcanism in the Long Valley area began about 3.2 m.y. ago with the widespread eruption of trachybasaltic/andesitic lavas, followed locally by less voluminous quartz latite domes and flows in the vicinity of the present site of Long Valley caldera. These precaldera mafic and intermediate lavas, probably from mantle and deep crustal sources, were precursors to development of a large shallow magma chamber from which most of the subsequent, more silicic Long Valley eruptions were derived. The rhyolite complex at Glass Mountain, exposed in the northeast wall and rim of Long Valley caldera, was built by early eruptions from this shallow magma chamber between 1.9 and 0.8 m.y. ago [*Metz and Mahood*, 1984]. Catastrophic rupturing of the roof of this chamber was accompanied by devastating Plinian and ash flow eruptions 0.73 m.y. ago. These eruptions caused collapse of the roof of the chamber to form Long Valley caldera and produced the extensive deposits of Bishop Tuff that surround it.

Almost immediately after caldera collapse, eruptions resumed on the caldera floor, resulting in the accumulation of thick rhyolite tuffs and the outpouring of hot, fluidal rhyolite flows. These early postcaldera rhyolites consist typically of aphyric to sparsely porphyritic obsidian, containing only a few percent phenocrysts of plagioclase, hypersthene, Fe-Ti oxides, and biotite (in decreasing abundance). Eruption of this early rhyolite persisted during a relatively short time span ($\leq$ 100,000 years) between 0.73 and 0.62 m.y. ago. These eruptions were accompanied by resurgent doming of the caldera floor which resulted in contemporaneous uplift and faulting of the early rhyolite tuffs and flows. As a consequence, the west central part of the caldera is now occupied by a broad, arched resurgent dome that is 10 km in diameter and 500 m high, and is transected by a 2-km-wide, complexly faulted, northwest-trending medial graben. During resurgence a caldera lake also was forming, so that the rising dome formed an island within it. Continued rise of the dome eventually caused the lake to overflow the topographically low southeast caldera rim, resulting in downcutting of the Owens River Gorge and gradual draining of the lake.

During the past 0.5 m.y. as the lake receded, eruptions were confined to the caldera moat peripheral to the resurgent dome. This moat rhyolite contrasts texturally and mineralogically with the early rhyolite in being coarsely prophyritic and typically containing large phenocrysts of plagioclase, sanidine, quartz, hornblende, biotite, and Fe-Ti oxides (in decreasing abundance). It is also characteristically pumiceous and was apparently quite viscous, as domes and flows are usually thick, steep-sided, and of limited extent. The moat rhyolite erupted during three distinct episodes, each comprising four or five individual domes or flows with associated pyroclastics and each localized in a different sector of the caldera moat. The eruptive episode occurred at about 0.5 m.y. ago on the north flank of the resurgent dome, about 0.3 m.y. ago in the southeast moat, and about 0.1 m.y. ago in the west moat. Thus the moat rhyolite episodes are separated by approximately 0.2-m.y. intervals and follow a clockwise succession

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around the resurgent dome. The cause of this 0.2-m.y. periodicity is poorly understood, but it may be related to the time required to develop sufficient vapor pressure in the magma chamber to drive eruptions [*Bailey et al.*, 1976] or to the periodic influx of basalt into the roots of the magmatic system, causing periodic thermal and convective rejuvenation of the shallower silicic chamber [*Bailey*, 1984].

During the latter part of the moat rhyolite sequence, between 0.25 and 0.05 m.y. ago, trachybasaltic and quartz latitic lavas began erupting in and near the west moat of the caldera. These mafic and intermediate lavas resemble the 3.2-m.y.-old precaldera mafic-intermediate sequence, but are less extensive, and are localized along a linear zone extending northward from the Devils Postpile through the west moat of Long Valley caldera to the north shore of Mono Lake. The largest concentration of these young mafic-intermediate lavas is in the western part of Long Valley caldera, where at least 30 cinder cones and other types of vents can be identified. Trachybasaltic flows from these vents inundated the western caldera floor, some flowing eastward around the resurgent dome, sending long tongues into the north and south moats. The largest accumulation of quartz latite lavas is on Mammoth Mountain, a large cumulovolcano on the southwest caldera rim; other large quartz latite domes occur within the southwest and northwest moat. These quartz latites, previously referred to as "rim rhyodacites" [*Bailey et al.*, 1976] and thought to be derived from deeper levels in the Long Valley magma chamber, are now considered along with the temporally associated trachybasaltic lavas to represent a new cycle of mafic-intermediate volcanism from a source deeper than the silicic Long Valley magma chamber, from which the early and moat rhyolites erupted (R. A. Bailey, unpublished data, 1985).

From 50,000 years ago, when the youngest summit dome on Mammoth Mountain erupted, to 550 years ago, no eruptions are known to have occurred in Long Valley. However, volcanism did continue further north at the Mono Craters where some 30 rhyolite domes and associated pyroclastics erupted between 40,000 and 600 years ago [*Wood*, 1979, 1983; *Sieh*, 1984]. Apparent southward extension of this activity 550 years ago resulted in eruption of the Inyo rhyolite domes in the northwest moat of Long Valley caldera as well as several phreatic explosion craters on the summit and south flank of Deer Mountain and on the north face of Mammoth Mountain [*Miller*, 1985]. Recently intensified seismicity and ground deformation in the area and numerous recent research studies indicate that magma persists at depth beneath Long Valley caldera and suggest the potential for future eruptions in the area [*Miller et al.*, 1982].

The rocks sampled in this paleomagnetic study include the postcaldera early rhyolites and moat rhyolites of Long Valley and the trachybasaltic and quartz latitic rocks of the younger cycle that occur in the west moat of the caldera. They span the period from 0.75 m.y. to 50,000 years ago.

METHODS

Samples for paleomagnetic analysis and potassium-argon age determinations were collected at the localities shown in Figure 1. The informal names given the rock units within the caldera by *Bailey et al.* [1976] are used throughout this report.

Three to eight oriented cores were obtained from each paleomagnetic sample locality. As volcanic domes are erupted, their carapaces may cool and become magnetized while magma is still being forced into their interiors. For this reason, rhyolite and rhyodacite domes were sampled at more than one outcrop, where possible, in order to detect any rotation of individual blocks that might have occurred after their magnetizations were acquired. Alternating field (af) demagnetization experiments were performed on one or more samples from each locality. Sites with anomalous components of magnetization were magnetically cleaned at a peak alternating field chosen on the basis of behavior during the af demagnetization experiments. Strong-field thermomagnetic curves were obtained in a nitrogen atmosphere with applied fields between 2000 and 6000 Oe. Weak-field susceptibilities were measured at 1000 Hz with a commerical bridge.

The mean remanent magnetization directions, after magnetic cleaning if needed, and associated statistical parameters are given in Table 1. Also included in this table are the statistical data for the rhyolite-obsidian flow near Whitmore Hot Springs (locality 43, Figure 1), for which only the magnetic polarity and K-Ar age were previously published [*Doell et al.*, 1966]. A summary of intensity, susceptibility, and Koenigsberger ratio grouped according to rock type is given by *Williams et al.* [1977].

For the K-Ar analyses, biotite, sanidine, or anorthoclase was separated when possible, using standard magnetic and heavy liquid techniques. Sanidine and anorthoclase were treated with a dilute HF solution to remove small bits of attached glass or other mineral fragments. Obsidian used for dating was unhydrated and not devitrified. For obsidian samples that contain phenocrysts the sample was crushed to between 2 and 4 mm, and pure glass fragments were handpicked for analysis. Basalt and andesite samples were examined in thin section to ensure that they met the usual criteria for whole rock K-Ar dating [*Mankinen and Dalrymple*, 1972] and then crushed to $-10+18$ mesh for Ar analysis and -200 mesh for K_2O analysis.

Argon was measured by isotope dilution mass spectrometry using extraction techniques, mass analysis, and data reduction procedures described by *Dalrymple and Lanphere* [1969]. Potassium was determined using the lithium metaborate fusion technique and flame photometry [*Suhr and Ingamells*, 1966; *Ingamells*, 1970].

The analytical data and calculated K-Ar ages are given in Table 2 and discussed briefly in appropriate sections below. Detailed discussion of most of the K-Ar ages and their relation to the geologic history of the caldera was presented by *Bailey et al.* [1976]. A summary of the ages, showing stratigraphic relations and relevant statistical data, is presented in Figure 2.

RATES OF GEOMAGNETIC SECULAR VARIATION

Differences in paleomagnetic directions recorded by the volcanic rocks from Long Valley caldera are almost entirely due to geomagnetic secular variation and can be used as an aid in refining the sequence of eruptions within the postcaldera episodes of volcanism. Because secular variation is associated with changes in the earth's nondipole field having periods of 100 to 1000 years, paleomagnetic methods are potentially useful in distinguishing age differences that are for too small to be resolved by radiometric dating methods. In order to be able to estimate the timing of eruptions we first need to determine the "normal" rate of secular variation at Long Valley. A record of the secular variation of the geomagnetic field at London and Paris has been kept for 400 years and shows an angular change of approximately 5° per 100 years. However, because this record covers a rather brief period of time and the amount of variation is latitude dependent, an estimate needs to be made of changes that can be expected over a

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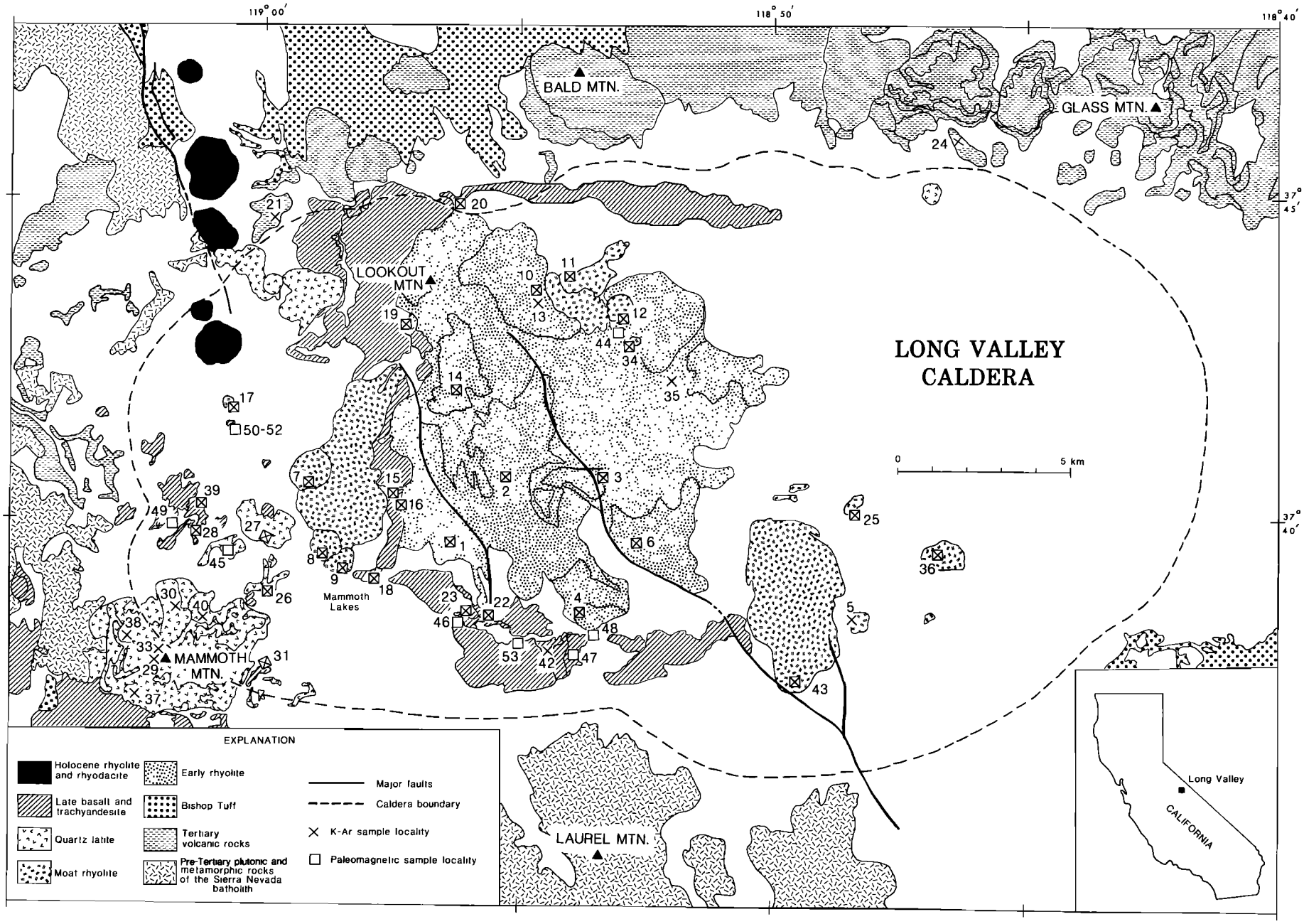


Fig. 1. Simplified geologic map of Long Valley caldera showing distribution of major rock types, eruptive units, and paleomagnetic and K-Ar sampling localities. The Bishop Tuff, the early and moat rhyolites, and the trachybasaltic and quartz latitic rocks are all Pleistocene in age. Unpatterned areas consist of alluvium, glacial and postcaldera pyroclastic deposits, and caldera fill.

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TABLE 1. Flow-Average Paleomagnetic Data for Volcanic Rocks From Long Valley Caldera, California

Flow Designation	Map Number (Figure 1)	Age, m.y.	H	Inclination, deg	Declination, deg	J_N	J_{CL}	N	R	K	α_{95}	σ	χ	Q	PLA	PLO
Late Basalts and Trachyandesites																
5L209	23	0.064	200	37.6	338.0	3.22	0.87	5	4.9829	234	5.0	5.3	19.9	3.1	64.8	116.3
5L233	53		100	37.9	332.3	1.72	1.47	8	7.9956	1604	1.4	2.0	18.3	1.8	61.0	124.6
5L352	31	0.086	100	25.8	017.7	1.57	1.13	8	7.9906	741	2.0	3.0	12.7	2.4	61.3	023.0
5L452	52		NRM	45.3	350.9	7.26		8	7.9789	331	3.1	4.5	10.7	12.9	76.7	098.8
5L460	51		NRM	47.9	345.2	4.51		8	7.9858	494	2.5	3.7	9.24	9.3	74.9	120.0
5L468	50		NRM	48.1	348.7	12.4		8	7.9452	128	4.9	7.2	8.07	29.2	77.3	112.0
5L344	28	0.095	100	48.4	005.8	4.04	2.87	8	7.9851	471	2.6	3.7	8.14	9.5	80.5	029.0
5L444	49		NRM	56.3	006.3	3.73		8	7.9864	515	2.5	3.6	16.9	4.2	84.9	338.5
5L334	39	0.103	200	58.5	358.9	1.78	1.27	5	4.9903	411	3.8	4.0	5.84	5.8	88.3	212.0
5L385	20	0.108	NRM	55.2	008.6	4.57		7	6.9877	487	2.7	3.7	6.55	13.3	82.8	345.2
5L360	48		200	60.2	340.2	3.37	0.82	8	7.9629	189	4.1	5.9	11.5	5.5	74.3	170.0
5L225	47		400	70.4	032.0	4.42	1.29	8	7.9524	147	4.6	6.7	5.13	16.4	62.5	282.7
5L201	46	0.116	NRM	58.6	021.7	2.38		8	7.9901	707	2.1	3.1	4.71	9.6	73.0	318.6
5L326	19	0.149	300	47.5	032.9	4.45	0.83	8	7.5368	15	14.7	21.0	8.36	10.1	n.d.	n.d.
5L217	22	0.153	50	57.5	014.8	4.49	4.14	8	7.9777	313	3.1	4.6	13.6	6.3	78.3	324.2
5L258	15	0.154	100	75.3	325.4	2.46	2.04	7	6.9911	673	2.3	3.1	6.02	7.8	n.d.	n.d.
5L266	16	0.155	200	84.2	002.1	2.62	1.74	8	7.8945	66	6.9	10.0	6.15	8.1	n.d.	n.d.
5L274	18	0.228	300	68.9	066.6	5.44	1.30	8	7.9625	187	4.1	5.9	13.0	8.0	n.d.	n.d.
Quartz Latites																
5L313	27	0.142	NRM	62.4	003.1	2.64		8	7.9733	262	3.4	5.0	5.54	9.1	83.6	261.1
5L321	26	0.148	NRM	58.4	356.8	6.31		5	4.9794	195	5.5	5.8	8.41	14.3	87.1	181.0
5L306	45		700	66.6	355.6	3.13	0.63	7	6.9188	74	7.1	9.4	6.69	8.9	78.2	226.9
Moat Rhyolites																
5L290	8	0.097	100	50.9	349.7	0.07	0.005	8	7.9484	136	4.8	7.0	0.47	2.8	79.6	118.3
5L282	9	0.106	500	48.1	358.8	0.36	0.002	4	3.9771	131	8.1	7.1	0.27	25.7	81.4	068.4
5L298	7	0.110	500	63.6	346.1	0.06	0.002	6	5.9623	133	5.8	7.1	0.69	1.8	77.2	191.4
5L420	17	0.115	50	65.7	336.3	0.24	0.19	8	7.9633	191	4.0	5.9	2.81	1.7	70.0	189.1
2C565	43	0.29	NRM	47.0	358.9	1.24		4	3.9796	147	7.6	6.7	1.76	13.4	80.5	067.0
5L492	36	0.329	200	66.5	347.0	0.27	0.21	5	4.9707	136	6.6	6.9	0.43	8.3	75.3	205.5
5L484	25	0.369	500	61.4	349.1	0.49	0.002	6	5.9496	99	6.8	8.1	0.44	3.8	80.4	184.7
5L401	11	0.481	400	45.8	014.0	0.51	0.17	8	7.9663	208	3.9	5.6	2.22	4.4	74.2	008.8
5L409	12	0.523	50	51.7	002.5	0.19	0.17	7	6.9550	133	5.3	7.0	1.32	2.8	84.3	039.4
5L476	34	0.527	400	50.1	017.5	0.34	0.18	8	7.9815	379	2.9	4.2	2.09	3.1	74.0	351.3
Early Rhyolites																
5L417	44		NRM	72.9	314.7	1.91		3	2.9984	1272	3.5	2.3	2.17	16.8	n.d.	n.d.
5L368	14	0.652	300	53.6	336.0	2.34	1.52	8	7.9351	108	5.4	7.8	2.14	20.8	70.3	147.9
5L377	2	0.678	NRM	52.1	005.9	2.49		8	7.9505	141	4.7	6.8	2.53	18.7	83.1	015.5
5L428	3	0.680	200	73.4	023.6	2.69	2.06	5	4.9806	206	5.3	5.6	3.11	16.5	63.7	268.6
5L393	10	0.685	NRM	70.6	007.0	0.34		8	7.9173	85	6.1	8.8	1.33	4.9	72.3	254.3
5L241	4	0.688	NRM	69.9	004.9	5.94		8	7.4682	13	15.9	22.5	1.22	93.2	n.d.	n.d.
5L249	1	0.694	500	67.2	010.4	1.60	0.15	6	5.9844	321	3.8	4.5	2.83	10.8	75.6	268.9
5L436	6	0.751	NRM	46.3	022.7	0.11		8	7.9418	120	5.1	7.4	0.28	7.8	68.5	352.4

H, peak demagnetizing field (Oe). For sites where af demagnetization did not change the direction of magnetization nor improve its accuracy, the original measurements (NRM, natural remanent demagnetization) are listed. Inclination is positive downward; declination is east of north; J_N and J_{CL}, geometric mean intensity of NRM and after magnetic cleaning, respectively ($\times 10^{-3}$ emu/cm^3); N, number of samples; R, resultant vector of N unit vectors; K, precision parameter [*Fisher*, 1953]; α_{95}, 95% confidence cone about average direction; σ, angular standard deviation of mean direction; χ, geometric mean susceptibility ($\times 10^{-4}$); Q, geometric mean in situ ratio of remanent to induced magnetization (Koenigsberger ratio); PLA, PLO, north latitude and east longitude, respectively, of VGP position; n.d., values not determined.

longer period of time in a particular area. *Champion* [1980] has compiled virtual geomagnetic poles (VGP's) for volcanic rocks erupted during the last 2000 years in various regions of the world and averaged the VGP's for 100 year intervals within each region. Calculating the inclination and declination of the magnetic vector that would be produced at Long Valley by each of the VGP's calculated by *Champion* [1980] for the western United States yields the record of secular variation shown in Figure 3.

It should be emphasized that the record shown in Figure 3 is a synthetic record of secular variation that is predicted from *Champion's* [1980] VGP data. The actual record could be somewhat different because the VGP data have been smoothed by averaging and because of variations that might exist within the region. Also, as will be pointed out in a later section, the Holocene seems to have been an unusually quiet period in terms of secular variation with the amplitude of the signal (the dispersion of VGP's) considerably smaller than it was during earlier periods. The possibility then arises that the rate of angular change may also have been anomalously low. We assume, however, that the synthetic record will yield at least a rough approximation of the actual secular variation at Long Valley during the Quaternary.

The rates of secular variation are quite variable. At times of rapid change, for example, between 400 and 500 A.D., or 1000 and 1100 A.D., sizeable differences in magnetic field directions are seen after a few tens of years. However, during periods of slower variation, approximately 300 years may be necessary to produce paleomagnetically significant differences in direction. The average rate of change at Long Valley for the past 2000

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TABLE 2. K-Ar Ages and Analytical Data for Volcanic Rocks From Long Valley Caldera, California

Sample	Map (Figure 1)	Rock Type	Material Dated	K_2O,* wt %	Argon			Calculated Age, 10^6 years
					Weight, g	$^{40}Ar_{rad}$, 10^{-12} mol/g	$\frac{100^{40}Ar_{rad}}{^{40}Ar\ total}$	
72G001	1	Aphryic rhyolite	Obsidian	5.10(2)	3.461	4.948	29.1	
					3.470	5.189	43.5	0.694 ± 0.017
72G002	2	Biotite rhyolite	Obsidian	5.24(2)	4.228	5.125	53.4	
					4.188	5.070	16.5	0.678 ± 0.028
72G003	3	Pyroxene rhyolite	Obsidian	5.25 ± 0.06(4)	3.870	5.199	16.7	
					4.075	5.131	34.6	0.680 ± 0.029
72G004	4	Pyroxene rhyolite	Obsidian	5.22(2)	4.426	5.312	34.7	
					4.442	5.057	57.0	0.688 ± 0.014
72G005	5	Hornblende-biotite rhyolite	Sanidine	10.96(2)	5.828	5.263	48.2	0.333 ± 0.010
72G006	6	Pyroxene rhyolite	Obsidian	5.02(2)	4.089	5.583	46.4	
					4.197	5.280	45.3	0.751 ± 0.016
72G007	7	Hornblende-biotite rhyolite	Sanidine	10.88(2)	12.182	1.724	64.4	0.110 ± 0.011
72G008	8	Hornblende-biotite rhyolite	Sanidine	11.24(2)	5.080	1.330	9.1	
					5.872	1.590	33.5	0.097 ± 0.006
72G009	9	Hornblende-biotite rhyolite	Sanidine	11.32(2)	6.755	1.735	38.8	
					6.112	1.744	34.2	0.106 ± 0.002
72G010	10	Aphryic rhyolite	Obsidian	5.07(2)	3.430	5.145	43.9	
					4.458	4.967	53.0	0.692 ± 0.014
72G011	11	Hornblende-biotite rhyolite	Sanidine	10.64(2)	7.160	7.440	40.7	
					6.544	7.293	40.4	0.481 ± 0.010
72G012	12	Hornblende-biotite rhyolite	Sanidine	10.45(2)	5.799	8.023	62.3	
					5.158	7.730	76.0	0.523 ± 0.011
72G013	13	Aphryic rhyolite	Obsidian	5.14(2)	4.932	4.992	54.5	
					5.384	5.036	41.6	0.677 ± 0.014
72G014	14	Pyroxene rhyolite	Obsidian	5.15(2)	3.955	4.958	57.4	
					3.908	4.725	34.5	0.652 ± 0.014
72G015	15	Basalt	Basalt	1.65(2)	10.578	0.3665	3.1	0.154 ± 0.072
72G016	16	Basalt	Basalt	1.66(2)	11.078	0.3716	8.8	0.155 ± 0.028
72G017	17	Hornblende-biotite rhyolite	Sanidine	10.63(2)	4.347	1.890	24.8	
					12.901	1.728	32.2	0.115 ± 0.003
73G001	18	Andesite	Andesite	2.13 ± 0.01(4)	9.102	0.6996	2.8	0.228 ± 0.082
73G008	19	Basalt	Basalt	2.00 ± 0.01(4)	8.379	0.4280	8.8	0.149 ± 0.016
73G009	20	Basalt	Basalt	1.63 ± 0.05(8)	14.187	0.2011	4.3	
					12.256	0.2708	13.3	0.108 ± 0.012
73G010	21	Andesite	Andesite	3.01 ± 0.01(4)	14.401	12.79	70.2	2.95 ± 0.09
73G012	22	Basalt	Basalt	2.26 ± 0.01(4)	7.784	0.3717	4.7	
					5.837	0.4824	5.5	0.129 ± 0.026
			Basalt	2.30 ± 0.04(4)	14.688	0.6170	6.1	
					9.254	1.080	5.8	0.214 ± 0.034
73G013	32	Basalt	Basalt	1.42 ± 0.01(4)	10.534	0.2120	1.6	0.104 ± 0.064
			Basalt	1.42 ± 0.01(4)	14.129	0.3126	1.2	
					15.280	0.3225	1.3	0.156 ± 0.086
73G014	23	Basalt	Basalt	1.82 ± 0.01(4)	12.806	0.1484	4.3	
					11.736	0.1945	3.5	0.064 ± 0.014
			Basalt	1.79 ± 0.01(4)	14.770	0.5990	2.1	
					15.034	1.134	5.7	0.373 ± 0.065
73G016	24	Biotite rhyolite	Sanidine	9.75 ± 0.26(4)	1.053	27.75	61.6	1.96 ± 0.06
73G017	25	Hornblende-biotite rhyolite	Sanidine	11.34 ± 0.03(4)	4.084	6.029	55.5	
					3.433	5.812	58.2	0.362 ± 0.008
			Biotite	8.93 ± 0.06(4)	1.337	6.645	4.5	0.517 ± 0.109
73G018	26	Rhyodacite	Biotite	8.60(2)	1.688	2.212	4.1	
					1.995	1.423	2.8	0.148 ± 0.031
73G019	27	Rhyodacite	Biotite	7.12(2)	1.897	1.460	2.6	0.142 ± 0.081
73G021	28	Basalt	Basalt	1.57 ± 0.01(4)	10.930	0.2245	4.8	
					11.928	0.2108	5.7	0.095 ± 0.013
73G022	33	Hypersthene-augite-biotite quartz latite	Sanidine	7.36(2)	3.168	1.095	9.2	
					3.029	0.5492	5.2	0.070 ± 0.012
			Biotite	8.50(2) ± 0.021	6.430	1.581	3.0	0.129 ± 0.065

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TABLE 2. (continued)

Sample	Map (Figure 1)	Rock Type	Material Dated	K_2O,* wt %	Argon: Weight, g	Argon: $^{40}Ar_{rad}$, 10^{-12} mol/g	Argon: $\frac{100^{40}Ar_{rad}}{^{40}Ar\ total}$	Calculated Age, 10^6 years
73G023	29	Rhyodacite	Sanidine	7.76(2)	2.251	0.5824	9.2	0.052 ± 0.010
			Biotite	8.58 ± 0.02(4)	4.807	1.090	2.1	
					3.054	1.397	2.8	0.097 ± 0.025
73G024	30	Rhyodacite	Biotite	8.22(2)	2.633	1.483	2.6	
					1.841	1.436	2.4	0.123 ± 0.035
73G025	31	Andesite	Andesite	2.63 ± 0.01(4)	9.418	0.3195	14.8	
					7.543	0.3656	7.1	0.086 ± 0.010
74G001	34	Hornblende-biotite rhyolite	Sanidine	10.62 ± 0.08(6)	2.419	8.380	24.4	
					5.343	7.833	60.7	0.527 ± 0.012
			Biotite	8.54 ± 0.11(6)	5.148	8.523	12.2	
					2.318	6.955	7.7	0.654 ± 0.041
74G002	35	Hornblende-biotite rhyolite	Sanidine	10.60 ± 0.08(4)	4.786	7.704	67.2	0.505 ± 0.015
			Biotite	8.48 ± 0.04(4)	5.447	8.873	8.9	0.727 ± 0.087
74G003	36	Hornblende-biotite rhyolite	Sanidine	9.72(2)	3.625	4.840	6.7	
					3.674	4.223	3.5	
					5.094	5.553	5.4	
					5.009	4.229	6.8	0.329 ± 0.023
74G004	37	Rhyolite	Biotite	8.12(2)	5.357	2.522	4.0	
					4.772	3.915	4.4	0.256 ± 0.046
74G005	38	Augite-biotite rhyodacite	Biotite	8.17(2)	4.520	1.905	2.7	
					4.473	2.967	5.0	0.215 ± 0.039
74G006	39	Dacite	Dacite	3.91 ± 0.03(4)	20.414	0.6650	7.8	
					19.704	0.5744	25.4	0.103 ± 0.009
74G009	40	Biotite rhyolite	Anorthoclase	2.69 ± 0.05(6)	13.032	0.8487	14.8	
					14.156	0.6249	12.7	0.212 ± 0.018
			Biotite	7.82 ± 0.08(4)	4.651	1.709	2.6	
					3.454	2.353	3.1	0.186 ± 0.038
74G010	41	Basalt	Basalt	1.55 ± 0.03(12)	14.828	1.890	5.1	
					20.000	1.497	4.1	
					20.111	1.651	2.9	0.729 ± 0.092
74G011	42	Basalt	Basalt	0.977 ± 0.014(12)	19.700	0.3191	5.3	
					18.992	0.3561	5.2	0.239 ± 0.029
509-27-6	43	Rhyolite	Obsidian	4.85 ± 0.03(4)	5.429	2.008	19.7	0.288 ± 0.031

Ages calculated using 1976 International Union of Geological Sciences (ICSU), Ottawa, Ont. constants [*Steiger and Jäger*, 1977]: Lambda B = 4.962×10^{-10} yr^{-1}, lambda E = 0.581×10^{-10} yr^{-1}, K-40/K-total = 1.167×10^{-4} mol/mol. Errors are estimates of analytical precision at the 68% confidence level for single measurements and of the standard error of the weighted means for replicate measurements. Means are weighted by the inverse of the variance.

*Mean and, where more than two measurements were made, standard deviation. Number of measurements is in parentheses.

years appears to be about 8° of arc per 200 years. Thus, as a first approximation, the fact that two or more volcanic units yield the same K-Ar age and paleomagnetic direction would be permissive evidence that the units could have been erupted within 100 years of each other. Similarly, units that yield the same K-Ar age but statistically different paleomagnetic directions must differ in age by more than 200 years.

EARLY RHYOLITES

Paleomagnetic and K-Ar samples were collected from 8 of the 12 early rhyolite domes and flows exposed on the resurgent dome. The average remanent magnetization directions for these units are shown in Figure 4. Five of the rhyolites sampled, localities 1, 2, 3, 4, and 10, yielded ages that are not significantly different from each other (Table 2 and Figure 2). K-Ar localities 10 and 13 are in the same flow, and the mean age for this rhyolite unit is 0.685 ± 0.008 m.y. The mean magnetization directions for four of these rhyolites (localities 1, 3, 4, and 10) are also quite similar, providing evidence that they could have erupted at about the same time. However, there is a large uncertainty in the direction determined for sample locality 4, where samples were collected from four small outcrops on the obsidian carapace of the rhyolite dome. The directions at each outcrop are somewhat different, showing that rotation of the individual blocks had occurred after cooling due either to extrusion mechanics or subsequent faulting. Thus, while the mean direction of all the outcrops coincides with the directions for localities 1, 3*a*, and 10, the agreement could be fortuitous.

Two outcrops were sampled at locality 3. The outcrop at locality 3*a* is large and well exposed and is probably undisturbed, as evidenced by the close agreement of its magnetization direction with the directions for localities 1 and 10, which are statistically similar in age. In contrast, the westernmost outcrop (3*b*) was small and poorly exposed, so only three samples were collected. As can be seen in Figure 4, this outcrop has apparently been rotated, approximately 35° to 40° toward the southwest.

The magnetization direction for the rhyolite at locality 2 is different from that for other rhyolite units of the same apparent age, indicating that it was erupted at a somewhat different

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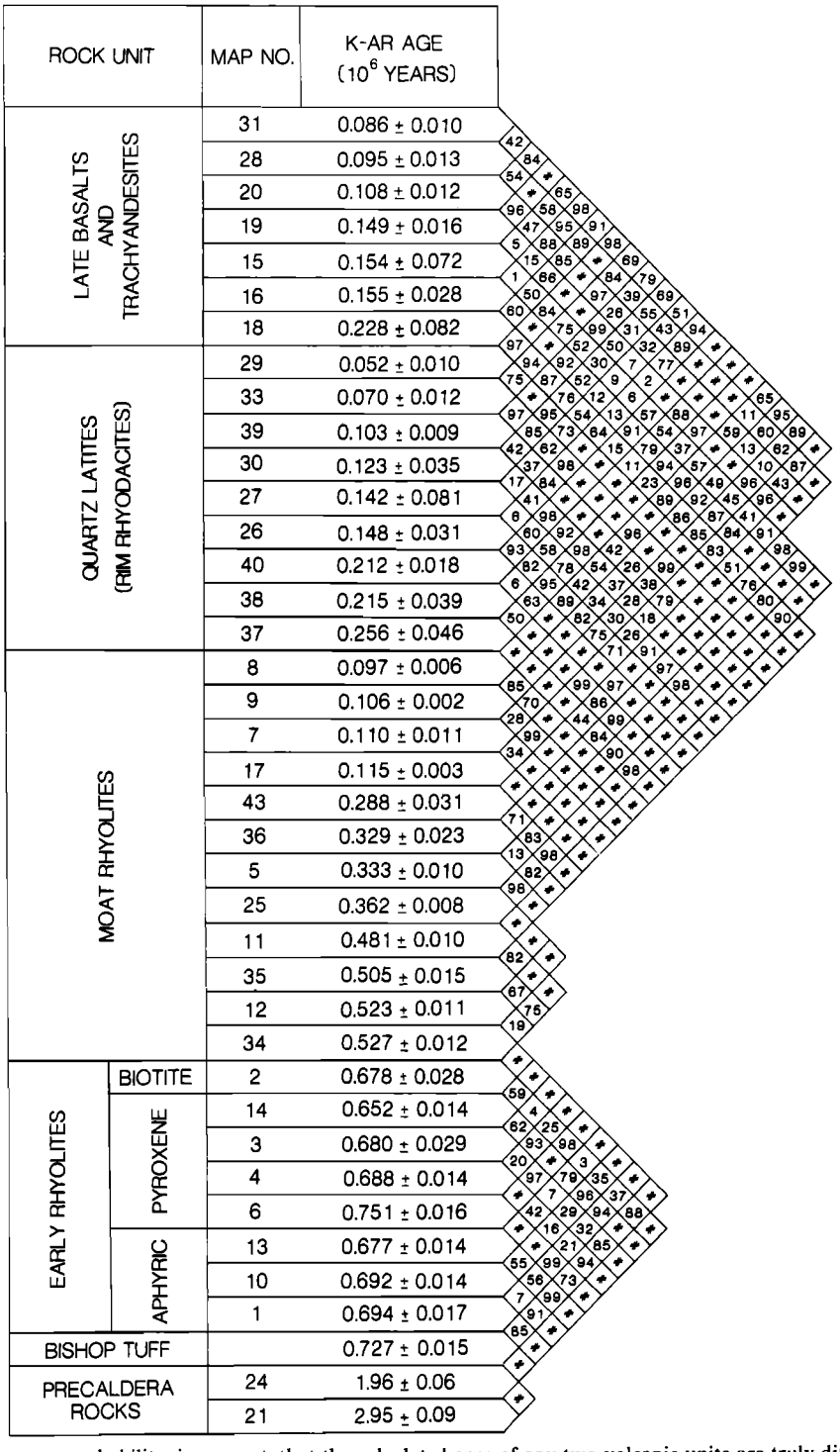


Fig. 2. Maximum probability, in percent, that the calculated ages of any two volcanic units are truly different and not the result of random errors in the analytical procedure. A probability of more than 99% is indicated by a pound sign. For six samples containing biotite and sanidine or anorthoclase the age plotted is that of the feldspar.

time. However, its age may differ by only a few hundred years, which would be unresolvable by radiometric dating methods. Another possibility is that this entire unit was tilted during resurgence, as has apparently occurred in other areas near the center of the structural dome [*Bailey et al.*, 1976]. Because of the wide geographic distribution of localities 1, 3, 4, and 10 about the central resurgent dome any appreciable radial tilting during resurgence should be reflected in the magnetic directions of these units if they are truly all the same age. The fact that there is little difference in magnetic direction suggests that differential tilting did not accompany uplift and that the tilting that has been observed is local in extent. For these four units to be of different ages and to have identical magnetization directions through either coincidence or random rotation seems highly unlikely.

Outcrops of the early rhyolite flow on the northeastern side of the resurgent dome are poorly exposed and very limited. For this reason, only 3 samples could be collected at locality

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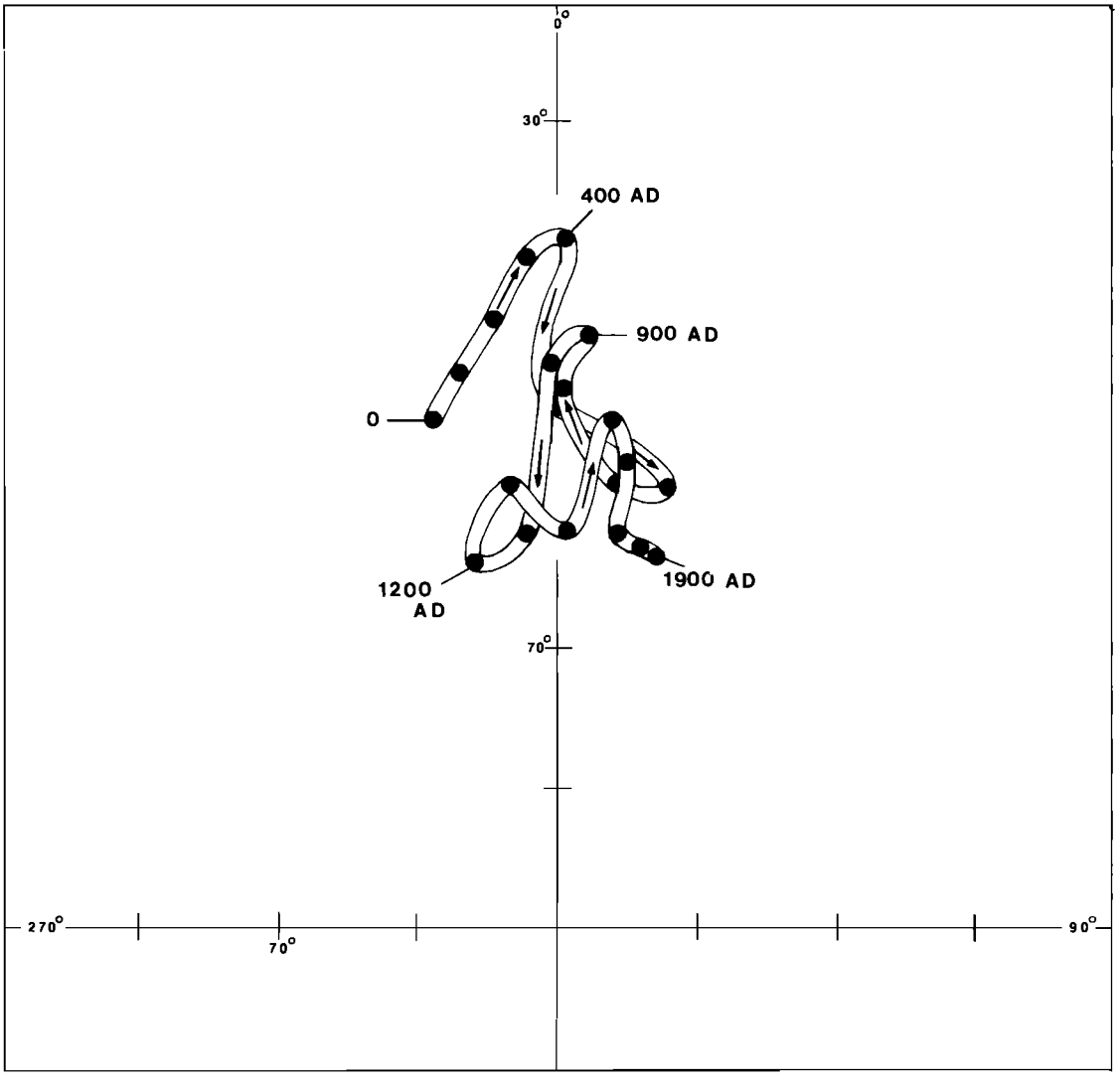


Fig. 3. Synthetic record of secular variation at Long Valley over the last 2000 years. Dots are directions produced by mean VPG's for each century in the western United States compiled by *Champion* [1980].

44 as a check on polarity. No significance should be attached to the magnetization direction for this locality other than that it is clearly of normal polarity.

Moat Rhyolites

The oldest group of moat rhyolites erupted about 0.5 m.y. ago from five centers, four of which are aligned along a NW trend on the northeastern flank of the resurgent dome. Four of the five were dated, but only three were suitable for paleomagnetic sampling. The K-Ar ages for localities 12 and 34 (0.523 and 0.527 m.y., respectively) are not significantly different, but their magnetic directions (Figure 5*a*) show that some time did elapse between eruptions of these two domes. The K-Ar age of the northwesternmost center, locality 11, is distinctly younger than the others in this group, and the similarity between the magnetic directions for localities 34 and 11 may be fortuitous. The older moat rhyolites, therefore, appear to have been erupted during at least three separate events.

A second group of five moat rhyolites was erupted at about 0.3 m.y. ago in the southeast moat of the caldera; four were dated, but only three were suitable for paleomagnetic sampling. Magnetic directions of samples from localities 25 and 36 are not significantly different (Figure 5*b*), although there appears to be some difference between their ages (Figure 2). The magnetic polarity and K-Ar age of the Hot Creek rhyolite-obsidian flow (locality 43) were previously published by *Doell et al.* [1966]. The magnetic direction and related statistics for this unit (A. Cox, written communication, 1965) are given in Table 1 and shown in Figure 5*b*. The age of the Hot Creek flow is distinctly younger than those of the other rhyolites of this group, which is also reflected in its magnetization direction.

A third group of four moat rhyolite domes erupted about 0.1 m.y. ago in the west moat of the caldera. The magnetic directions for these four domes (Figure 5*c*) suggest that the domes were erupted in pairs, 17 and 7, and 8 and 9. There is very little difference in direction within each pair, although some difference is indicated by the K-Ar ages. On the basis of the K-Ar ages, localities 8 and 9 belong to the younger group. The magnetic direction for locality 9 is not as well defined as those for the other domes in this group due to lightning effects which could be completely removed in only four of the eight samples collected. However, the direction defined by these four samples indicates that this locality is more closely related in time to locality 8, which seems reasonable because these two domes are immediately adjacent to each other (Figure 1).

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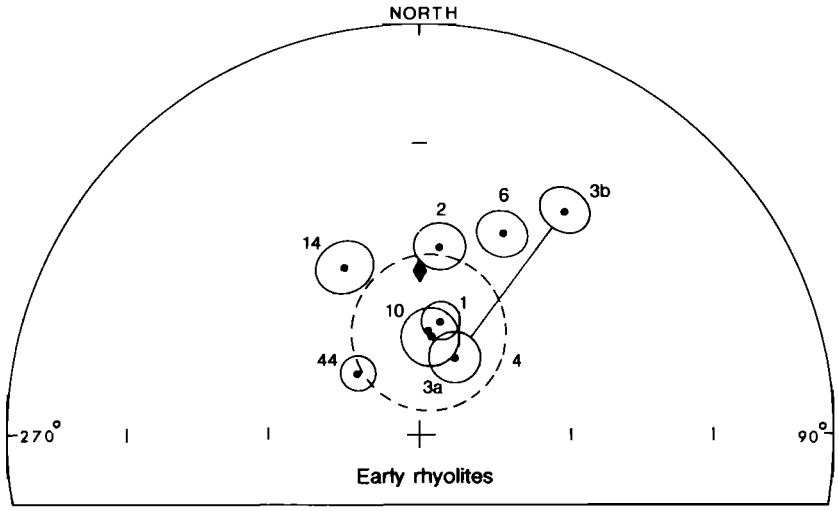


Fig. 4. Average remanent magnetization directions after magnetic cleaning for the early rhyolites. Ovals, 95% confidence limits about the mean directions. Diamond, direction produced at Long Valley by an axial dipole field. All directions on lower hemisphere of an equal-area projection.

MODE OF ERUPTION

In the sections that follow, it will be shown that the 0.1-m.y.-old moat rhyolites were accompanied by many other eruptions in the west moat of the caldera. However, here only the moat rhyolites are considered, since they are genetically related and probably controlled by the same sets of ring fractures. Because all the moat rhyolites typically show magnetic directions close to the average dipole field direction for the area, it is difficult to use them to differentiate between individual eruptions. In fact, the possibility that eruptions within each group of moat rhyolites occurred over a much shorter time span than that indicated by the K-Ar ages cannot be precluded solely on the basis of magnetic directions. Similarly, within each group it is difficult to distinguish increasingly small time differences with the K-Ar determinations. However, if the results of both methods are considered, the paleomagnetic results support the impression given by the K-Ar ages that the eruptions tended to be episodic even within each group of rhyolites. The rhyolites at localities 12 and 34 in the 0.5-m.y.-old group probably erupted at somewhat different times even though the ages are the same within analytical error. The angular difference between their magnetic directions is only 9.6°, however, so the time involved need not be more than a few decades. Thus, in this group, one eruptive episode (12 and 34) at 0.52 m.y. seems to have been followed some 40,000 years later by eruption of the flow at locality 11. The dome at locality 35 may or may not represent an intermediate event.

Data for the 0.3-m.y.-old group of moat rhyolites could be interpreted in two ways. One episode may include events at localities 5, 25, and 36 about 0.34 m.y. ago, followed 50,000 years later by the flow at locality 43. Perhaps more likely, the flow at locality 25 erupted at 0.37 m.y. was followed by eruptions at localities 5 and 36 at 0.33 m.y., ending with that at locality 43 after another 40,000 years. The eruptive sequence of the third (0.1 m.y. old) group of moat rhyolites is more obvious with the eruptions at sites 17 and 7 occurring about 0.112 m.y. ago followed some 10,000 years later by those at sites 8 and 9. However the data are interpreted, it is apparent that the eruptions did not occur randomly within the main dome-building episodes. Within each 0.2-m.y. episode, the eruptions seem to have occurred in discrete intervals, or pulses, each separated by some tens of thousands of years.

QUARTZ LATITES (RIM RHYODACITES)

Sample localities described in this report are confined to the largest of the quartz latite areas in the southwest part of the caldera. Only three localities were sampled for paleomagnetic study, and these are located in the caldera moat. Additional K-Ar samples were collected from six units on Mammoth Mountain, a massive and complex cumulovolcano on the caldera rim. The K-Ar determinations (Table 2) suggest episodic eruption of quartz latite in a manner similar to that of the moat rhyolites and early rhyolites. This impression is reinforced by the coincidence of magnetization directions for units at localities 26, 27, and 45 (Figure 6). Localities 26 and 45 are in different parts of the same unit, and their magnetization directions are not statistically different at the 95% confidence level using the method of *McFadden and Jones* [1981]. K-Ar ages for localities 27 and 45 are nearly identical, as are their magnetization directions. The close agreement in magnetic direction between the three localities is at least suggestive of eruption at about the same time.

Bailey et al. [1976] considered the biotite ages more reliable than the plagioclase ages for rocks from Mammoth Mountain. Petrographic studies showed that the plagioclase crystals had complex histories and probable xenocrystic cores. K-Ar results from coexisting biotite and plagioclase showed that the plagioclase ages were more variable and consistently older. Additional determinations on coexisting sanidine and biotite now show that the biotite ages are also somewhat in error. For rocks this young, sanidine might be expected to yield more reliable ages because it commonly contains less atmospheric argon than biotite. Moreover, there is evidence for some extraneous ^{40}Ar in other biotites from Long Valley. Biotites from moat rhyolite localities 25, 34, and 35 yield ages approximately 30% older than those on coexisting sanidine (Table 3). For the younger rocks on Mammoth Mountain, this effect is even more pronounced with biotite ages approximately 45% older than sanidine ages of rocks from localities 29 and 33. If this relationship holds for other caldera rocks as well, the ages for the quartz latites at other localities may actually be closer to the lower limits of their uncertainty ranges. This would place the quartz latites at localities 26 and 27 closer to 0.100 m.y. in age, a possibility that will be considered again in a later section.

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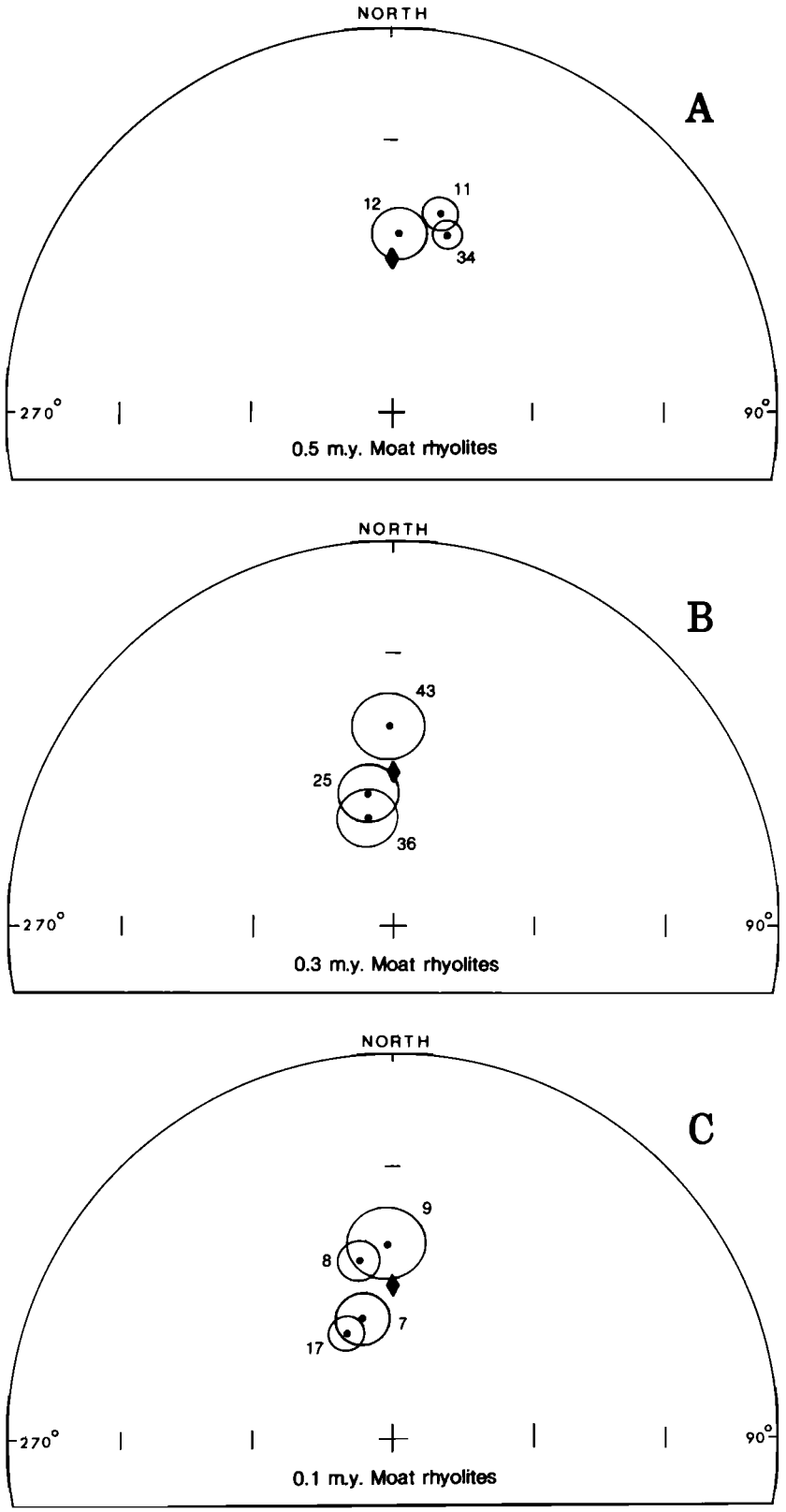


Fig. 5. Average remanent magnetization directions after magnetic cleaning for (*a*) 0.5-m.y., (*b*) 0.3-m.y., and (*c*) 0.1-m.y. groups of moat rhyolites. Symbols as for Figure 4.

LATE BASALTS AND TRACHYANDESITES

Magnetic directions for these mafic units are shown in Figure 7*a*. Four of the basaltic flows along the western margin of the resurgent dome are unquestionably of normal polarity (Figure 7*b*), but their directions were not used in the calculation of a mean VGP for the caldera. The flow at site 19 was sampled at two outcrops, both of which were somewhat jumbled, due either to frost action or recent faulting. While indi-

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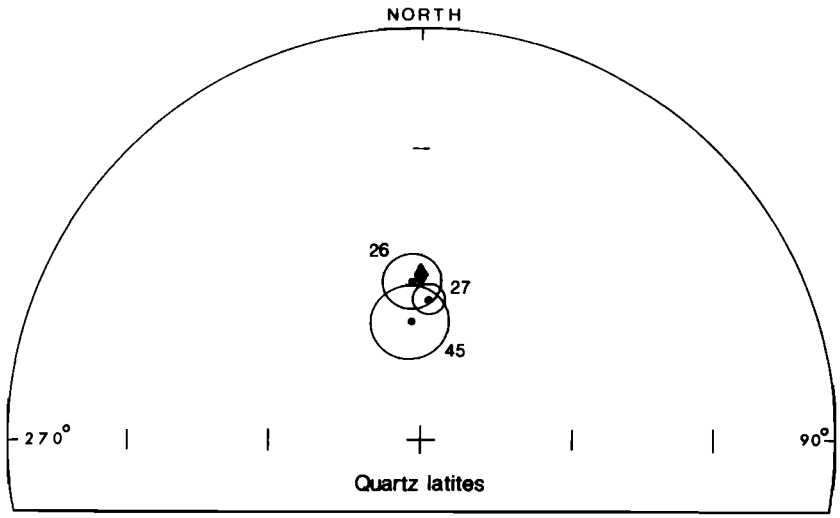


Fig. 6. Average remanent magnetization directions after magnetic cleaning for the postcaldera quartz latites. Symbols as for Figure 4.

vidual samples appear to reach stable end points during af demagnetization, little agreement between samples was found, which indicates that many of the blocks have been rotated with respect to each other.

Magnetization directions determined for localities 15, 16, and 18 (Figure 7*b*) have anomalously steep inclinations. Because the flow at locality 18 is significantly older than those at localities 15 and 16 and because the latter two flows also appear to have erupted at different times based on their magnetization directions, it is surprising that they should all have such unusual directions. A possibility is that the structural block containing these three sites has been tilted (approximately 20°) toward the north or northwest. While faults bounding this structural block have not been recognized, there is some indication that the flows may have been erupted onto an unstable surface which has subsequently slumped, tilting the flows toward the north. Alternatively, they may be tilted by late uplift of the resurgent dome.

The south moat of the caldera is a broad area of low relief covered by thick growth of sage. Paleomagnetic directions of the localities sampled (Figure 7*c*) agree with the succession mapped by *Bailey* [1974] and show that considerable time elapsed between some of the eruptions. Localities 23 and 53 were thought either to be from the same eruption or to be very closely related in time [*Bailey*, 1974], and this suggestion is substantiated (Figure 7*c*). These magnetization directions are far enough removed from the dipole field direction that they are unlikely to be the same through coincidence.

Sample localities 22 and 46 are in flows that underlie and overlie, respectively, the Casa Diablo till of *Curry* [1971]. On the basis of single K-Ar determinations on the underlying and two overlying flows, *Curry* [1971] assigned an age of about 0.4 m.y. to the Casa Diablo till. Age determinations by *Bailey et al.* [1976] showed the till to have an age between 0.062 m.y. and 0.126 m.y. and thus to be possibly equivalent in age to the Mono Basin till of *Sharp and Birman* [1963]. However, the reproducibility of the argon measurements for the two basalt flows [*Bailey et al.*, 1976] was poor, and a new sample from the original collection was prepared for analysis from the upper and lower flows as well as two samples from the middle flow. For samples from all three flows the K_2O contents of the duplicate preparations are in good agreement, but the argon contents are not. The latter differ by as much as an order of magnitude. Sample 74G010 is 0.729 ± 0.092 m.y. compared with a mean age of 0.122 ± 0.054 for 73G013 from the same flow. The upper flow (73G014) contains visible granitic inclusions. This fact, plus the large variation in $^{40}Ar_{rad}$ contents, indicates that all three flows probably contain extraneous argon, and so none of the ages can be considered reliable. The age of the Casa Diablo till, however, is still constrained by the youngest apparent age of the underlying flow. On this basis, the age of the Casa Diablo till is less than 0.126 m.y.

WEST MOAT VOLCANISM

Eruptions in the west moat of the caldera probably began about 0.23 m.y. ago with the eruption of trachyandesite (locality 18) and continued until 550 years ago with the eruption of the youngest of the Inyo rhyolite domes [*Miller et al.*, 1982; *Miller*, 1985]. One of the most active periods of volcanism in Long Valley occurred about 0.1 m.y. ago when a variety of rock types ranging in composition from basalt through rhyolite erupted in the west moat. Magnetization directions of 14 units sampled are shown in Figure 8*a*. All directions are tightly grouped and show little of the usual dispersion due to secular variation. This suggests that all eruptions occurred within a short period of time or that they were grouped in episodes of multiple eruptions. Flows sampled at localities 39, 49, and 28 (Figure 7*a*) are in stratigraphic sequence. When all directions are plotted without their confidence limits, the change in direction recorded by these three flows (solid circles, Figure 8*b*) defines an age trend on which the other units sampled could fall. All dated units fit within this trend, with Deer Mountain (locality 17) being one of the oldest eruptions and the rhyolite dome at locality 8 (Knolls dome) among the youngest. The directions of the 3 quartz latites, localities 26, 27, and 45 (triangles, Figure 8*b*), also fit within this age trend, suggesting but not proving that their ages may be nearer 0.1 m.y. than 0.15 m.y. as discussed earlier. The four directions at the tip of the arrow (Figure 8*b*) belong to the rhyolite dome at locality 8 and to the three andesite flows sampled in south Inyo crater (localities 50, 51, and 52). This crater is the southernmost of three phreatic explosion craters on the south flank of Deer Mountain that formed at about 550 years B.P. [*Miller*, 1985]. No primary lava was ejected with these ex-

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TABLE 3. Magnetic Properties of Samples

Map (Fig. 1)	Sample	J_{NRM}	χ	MDF	J_{rs}	J_s	J_{rs}/J_s	T_c
				Late Basalts				
23	211	3.700	26.700	150	0.2980	2.020	0.148	537
53	236	1.680	18.000	250	0.4270	2.200	0.194	565
31	353	0.823	12.700	175	0.0710	0.938	0.076	509
52	455	5.520	9.350	300	0.2060	1.250	0.165	540
51	461	4.580	9.140	425	0.1560	0.994	0.157	544
50	473	25.700	4.660	450	0.1570	0.670	0.234	(250), 400
28	345	4.110	9.070	400	0.1450	0.955	0.152	565
49	447	2.690	17.300	175	0.1810	1.640	0.110	532
39	335	1.300	9.200	125	0.0856	0.717	0.119	500
20	392	3.490	6.480	375	0.1930	0.960	0.201	570
48	364	2.990	10.100	75	0.0562	0.655	0.086	158
47	227	3.280	6.010	150	0.0858	0.585	0.147	190, 553
46	206	1.870	4.230	400	0.0532	0.497	0.107	565
19	328	5.350	9.520	50	0.1080	0.836	0.129	228, 556
22	219	4.410	15.500	200	0.1830	1.440	0.127	559
15	261	2.410	7.520	200	0.1800	0.862	0.209	557
16	271	3.040	6.980	350	0.2330	1.130	0.206	569
18	281	6.870	17.500	100	0.2390	1.650	0.145	(539), 565
				Quartz Latites				
27	315	3.130	7.300	350	0.1240	0.835	0.149	(535), 560
26	325	4.350	6.800	n.d.	0.1230	0.639	0.192	(521), 555
45	307	4.870	7.460	100	0.0214	0.489	0.044	525
				Moat Rhyolites				
8	290	0.131	n.d.	<100	0.0035	0.076	0.046	(534), 552
9	285	0.782	0.111	125	0.0021	0.067	0.031	(503), 541
7	301	0.347	n.d.	<100	0.0096	0.168	0.057	564
17	420	0.239	2.640	200	0.0123	0.286	0.043	(523), 550
43	566	1.360	1.940	500	0.0301	0.257	0.117	542
36	496	0.264	0.222	475	0.0030	0.036	0.083	556
25	488	3.020	n.d.	50	0.0065	0.103	0.063	538
11	401	0.293	2.050	350	0.0263	0.245	0.107	551
12	416	0.252	0.864	n.d.	0.0067	0.108	0.062	(528), 561
34	477	1.130	n.d.	75	0.0100	0.189	0.053	532
				Early Rhyolites				
44	418	3.230	2.330	525	0.0594	0.280	0.212	558
14	376	2.810	2.660	325	0.0442	0.258	0.171	540
2	377	3.790	2.770	350	0.0566	0.328	0.173	540
3	431	2.740	3.760	300	0.0632	0.345	0.183	536
10	399	0.302	0.968	350	0.0239	0.152	0.157	564
4	245	2.620	1.400	425	0.0271	0.201	0.135	535
1	255	2.390	2.780	150	0.0443	0.258	0.172	556
6	436	0.345	0.109	400	0.0037	0.025	0.149	560

J_{NRM}, intensity of NRM ($\times 10^{-3}$ emu/cm^3); χ, susceptibility ($\times 10^{-4}$); MDF, median demagnetizing field (Oe); J_{rs}, saturation remanence (emu/g); J_s, saturation magnetization (emu/g); T_c, Curie temperature (deg C). Curie temperatures in parentheses determined by extrapolation (see text); n.d., values not determined.

plosions, and the sampled flows predate the craters. While the age of these flows is uncertain and they might coincidentally have the directions shown, the fact that all other units sampled in the west moat fit in this tight grouping of directions suggests that these flows could be very close in age to the Knolls dome at locality 8.

It is unlikely that these west moat eruptions could have occurred continuously over as short an interval of time as would be indicated by direct comparison of their magnetization directions with the secular variation rates shown in Figure 3. The K-Ar ages indicate a period of about 20,000 years for them, and taking the uncertainty in the ages into account, a period of at least 10,000 years seems to be required. A directional change such as indicated in Figure 8*b* would suggest a period of a few hundred years at most unless the rates of secular variation were unusually slow at about this time. An alternative explanation is that many closely spaced eruptions occurred at discrete time intervals such as suggested by Figure 8*c*. In this interpretation the eruptive episodes may be separated by approximately 10,000 years, with the units sampled at localities 9 and 28 possibly belonging to another episode about 0.10 m.y. ago. The quartz latites may belong to an episode as shown in Figure 8*c*, or, more likely, at least two belong to the episode at 0.105 m.y. as shown by the means plotted in Figure 7*b*, and the third, to the 0.115-m.y. group. If the simple progression of magnetization directions in Figure 8*b* accurately represents secular variation, then the rate of variation would necessarily have been between 10 and 100 times slower than during the past 2,000 years. The fact that the discrete episodes also appear to define a progression is only coincidental because the magnetic field direction would have returned repeatedly to points near the axial dipole field direction over a 20,000 year interval. Whatever method is used to interpret the eruptive history of this part of the caldera, it is apparent that many eruptions occurred over a very short period of time.

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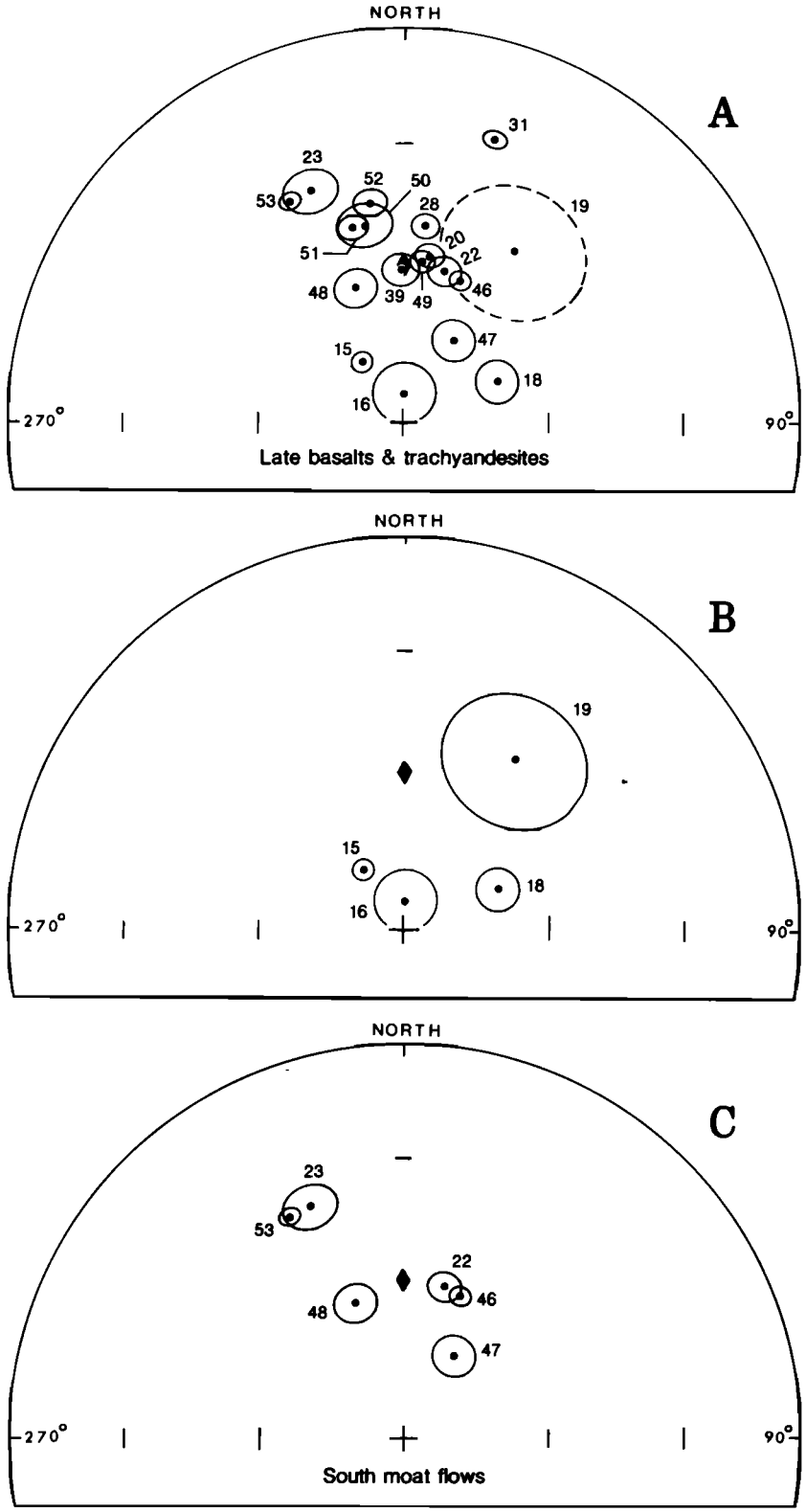


Fig. 7. (*a*) Average remanent magnetization directions after magnetic cleaning for all late trachybasalts and trachyandesites sampled. (*b*) Directions for flows thought to have been rotated. (*c*) Directions for reliable flows in the south moat of the caldera. Symbols as for Figure 4.

MAGNETIC PROPERTIES

Thermomagnetic analyses and measurement of rock magnetic properties were made on one specimen from each locality sampled. Results of these measurements are given in Table 3. Saturation remanence (J_{rs}) and saturation magnetization (J_s) were measured on the same sample used for thermomagnetic experiments. If the magnetic oxides are single domain, J_{rs}/J_s ranges from 0.50 to 0.87, while values between 0.1 and 0.5 represent either pseudosingle domains or mixtures of

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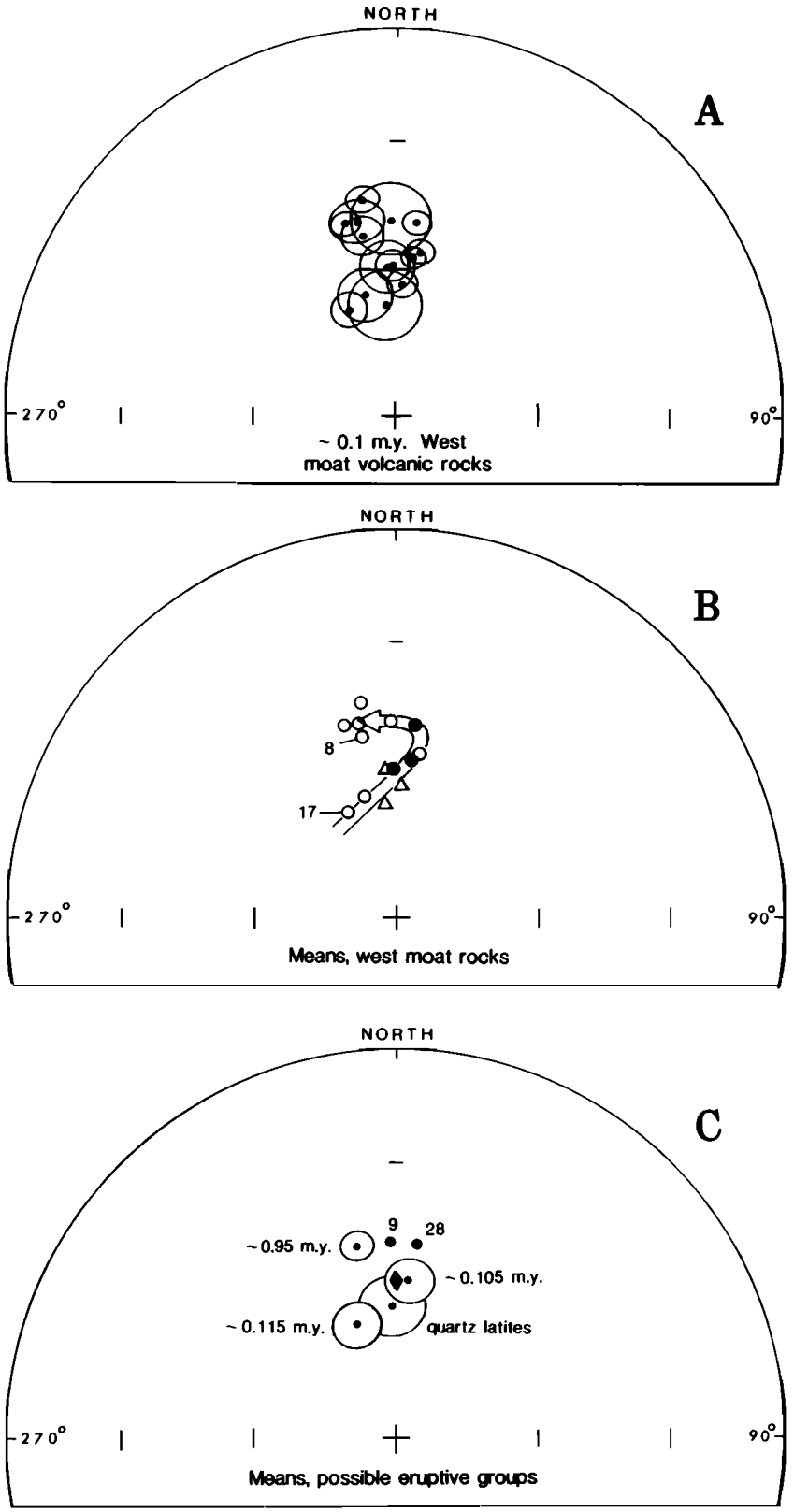


Fig. 8. (*a*) Average remanent magnetization directions and associated confidence ovals for all units sampled in the west moat of the caldera that are approximately 0.1 m.y. old. (*b*) Mean directions of all units in Figure 8*a* showing apparent change in direction with decreasing age in direction of arrow. Triangles, means of quartz latites; solid circles, flows in stratigraphic sequence. (*c*) Mean directions of possible groups of eruptions separated by approximately 10,000 years (see text). Other symbols as for Figure 4.

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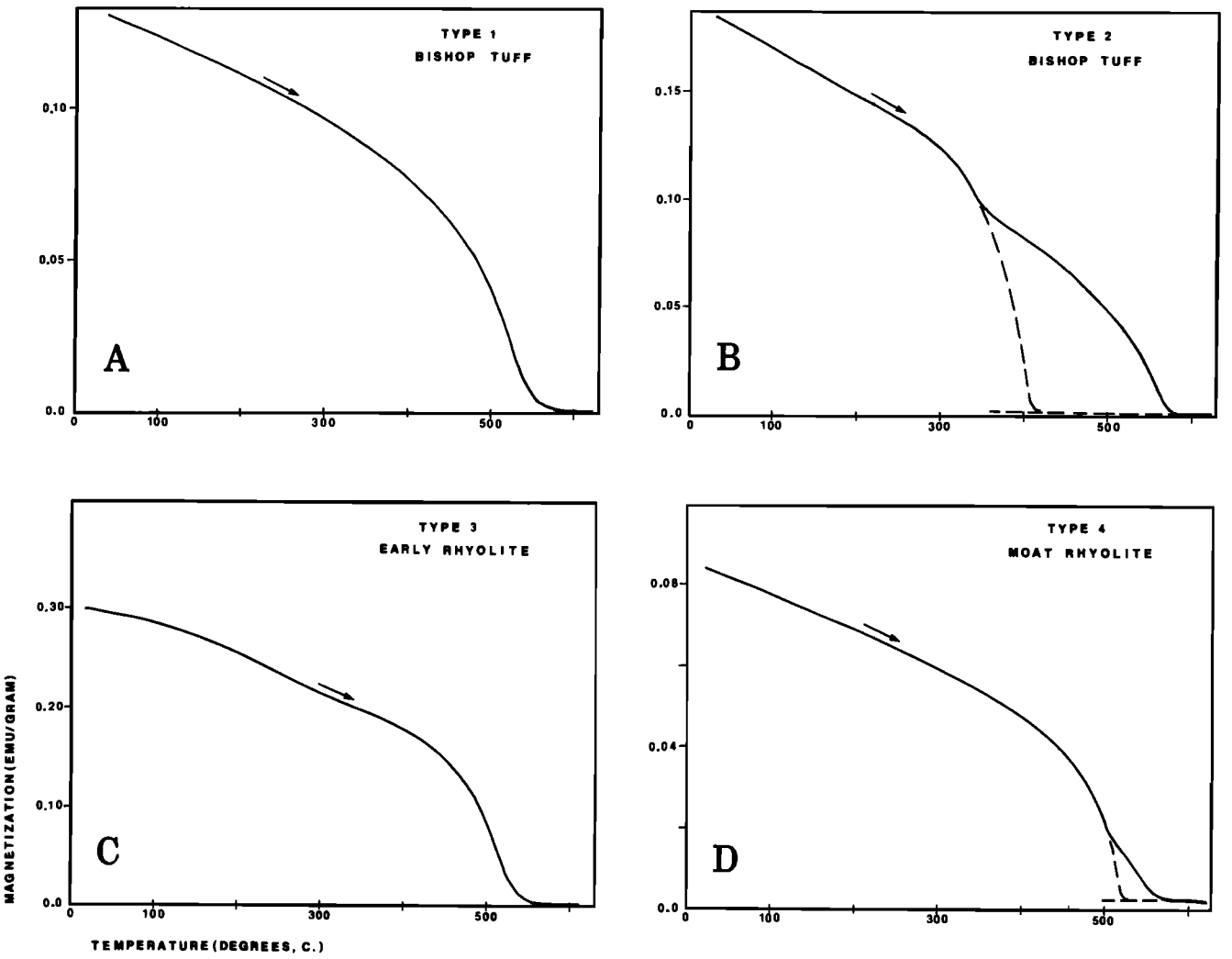


Fig. 9. Four types of strong-field thermomagnetic heating curves commonly observed for silicic rocks from the Long Valley magma chamber. All samples heated at 10°C/min. Dashed lines show extrapolation method of obtaining original Curie temperature (see text).

domain types [*Stacey and Banerjee*, 1974]. The measured ratios in these rocks are all less than about 0.2, which could indicate that fairly large pseudo single domain grains predominate. If, however, most of the magnetic grains in the Long Valley rocks contain magnetite/ilmenite intergrowths, J_{rs}/J_s for single-domain particles could be reduced to 0.30 due to large magnetic interactions [*Davis and Evans*, 1976]. A few samples, particularly those from the moat rhyolites, have ratios less than 0.1, indicating the exlusive presence of multidomain grains. However, this domain structure does not appear to be reflected in a lower magnetic stability of NRM. Although the median demagnetizing field (MDF) is low for some of the moat rhyolites having a low J_{rs}/J_s ratio, the MDF in these particular units is a reflection of the removal of large anomalous components of magnetization that are probably the result of lightning strikes.

Four types of strong-field thermomagnetic curves are commonly observed for silicic rocks related to the Long Valley magma chamber, and these are shown in Figure 9. Only the heating curves are shown for clarity. A description of the thermomagnetic curves for samples of the Bishop Tuff was given by *Williams et al.* [1977]. Inflections in the heating curves for samples of Bishop Tuff were found at temperatures of approximately 620°, 560°, and 350°C, which were thought to be due to the presence of ilmenohematite, titanomagnetite, and titanomaghemite, respectively. A comparison of some of the Bishop Tuff curves will be made with those of younger eruptive rocks from the Long Valley chamber. Ilmenohematite was detected in only a few samples of Bishop Tuff and will not be discussed further.

Type 1 thermomagnetic curves (Figure 9*a*), which indicate the presence of a single ferrimagnetic phase, are fairly common for all rock types analyzed. This phase may be a primary low-Ti titanomagnetite or may result from high-temperature deuteric oxidation of a primary Ti-rich titanomagnetite to form a low-Ti titanomagnetite containing ilmenite lamellae.

The lower temperature inflection point in the type 2 magnetization curves exhibited by the Bishop Tuff samples (Figure 9*b*) was previously thought to be due to the presence of titanomaghemite [*Williams et al.*, 1977]. Some samples from the Lava Creek tuff in the Yellowstone region exhibit very similar heating curves, which *Reynolds* [1977] also attributed to the presence of titanomaghemite or maghemite. Type 3 magnetization curves shown by samples of andesite and dacite from the Clear Lake volcanic field [*Mankinen et al.*, 1981] are also similar to the type 2 Bishop Tuff curves. *Mankinen et al.* [1981] suggested that this type of curve could be due to the presence of titanomaghemite or could indicate the presence of

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TABLE 4. Paleomagnetic Data for Combined Localities

Localities Combined	Inclination, deg	Declination, deg	N	R	K	α_{95}	σ	PLA	PLO
23, 53	37.9	334.5	13	12.9691	388	2.1	4.1	62.5	121.6
50, 51, 52	47.1	348.3	24	23.8946	218	2.8	5.5	76.5	110.5
26, 27, 45	62.9	359.0	20	19.8342	115	3.1	7.6	83.3	234.7
8, 9	50.0	352.8	12	11.9085	120	4.0	7.4	81.0	104.0
7, 17	64.9	340.7	14	13.9143	152	3.2	6.6	73.1	189.9
25, 36	63.7	348.2	11	10.9094	110	4.4	7.7	78.3	196.0
1, 3, 10	70.4	011.9	19	18.8525	122	3.1	7.3	71.4	263.1

See Table 1 for explanation.

two stoichiometric titanomagnetite phases. Magnetic measurements and petrologic studies of the Bishop Tuff now indicate that the presence of two titanomagnetite phases is the most likely explanation for this type of thermomagnetic curve.

As a rock containing titanomaghemite is heated, the titanomaghemite spontaneously inverts into its various breakdown products (commonly Ti-poor titanomagnetite and ilmenite), and the magnetization upon cooling is typically many times stronger than before heating due to the formation of new magnetite. However, type 2 curves do not show this behavior, and the cooling curves retrace only the portion of the heating curve that corresponds to the magnetic material with the highest Curie temperature. This change occurs whether the samples are heated in air or in vacuum, and the heating must convert the material having the lower Curie temperature into nonmagnetic or very weakly magnetic components. If two titanomagnetite phases are present, they probably result from cooling that was too rapid to allow complete high-temperature deuteric oxidation of an originally homogeneous titanomagnetite to Ti-poor titanomagnetite and ilmenite. Perhaps it is significant that the samples we have analyzed that show a large portion of the low Curie temperature material were all collected from one of the earliest (and coolest) of a succession of eruptions of Bishop Tuff [*Hildreth*, 1979]. One way to obtain an estimate of Curie temperature of the presumed Ti-rich phase is to extrapolate the continuation of the low-temperature part of the heating curve past the inflection point until it intersects the paramagnetic part of the curve as shown by the dotted lines in Figure 9*b*. Admittedly, this method is somewhat subjective, but it should yield an approximate Curie temperature for the original, unoxidized titanomagnetite. For the Bishop Tuff, this method yields Curie temperatures of slightly greater than 400°C for all samples that show this inflection. Assuming that this Curie temperature is representative of the titanomagnetite and using the temperature contours on the ulvöspinel-magnetite-hematite-pseudobrookite quadrilateral given by *Grommé et al.* [1979], an ulvöspinel content of not more than 30% is indicated. This estimate is in excellent agreement with *Hildreth* [1979], who found that the ulvöspinel content of microphenocrysts of unoxidized, unexsolved titanomagnetites from the Bishop Tuff range from 25.3 to 28.0 mol %.

The magnetic mineralogy of the early rhyolites is similar to that of the Bishop Tuff, which might be expected because they are closely related in time and may have tapped similar levels in the magma chamber. Samples from localities 2, 4, and 44 have single Curie temperatures and yield well-defined type 1 heating curves. Samples from the other localities probably originally had two titanomagnetite phases, as did the Bishop Tuff. However, the amount of Ti-rich titanomagnetite in these samples is either much less than that in the Bishop Tuff or occurs as a range of titanomagnetite populations. Consequently, the inflection point can no longer be clearly seen. Heating curves from these samples show only a slight depression at about 300°C (Figure 9*c*), and no extrapolation of the original Curie temperature is possible. Curie temperatures determined for these samples (Table 4) are only for the Ti-poor titanomagnetite phase.

The moat rhyolites and quartz latites again show either nearly coincident type 1 heating and cooling curves or contain two stoichiometric titanomagnetite populations. Using the extrapolation method as described for the Bishop Tuff, the lower Curie temperatures for these rocks average 525° (type 4, Figure 9*d*; Table 4). Plotting this value on the compositional diagram shows that the titanomagnetites in most of the later eruptive rocks have a considerably lower titanium content (~10%) than either the Bishop Tuff or the early rhyolites. Only the moat rhyolites sampled at localities 25 and 36 in the eastern part of the caldera show the type 3 curves commonly seen in the early rhyolites.

The late basalts and trachyandesites show a magnetic mineralogy that is typical of basaltic rocks. A sample from the basalt flow at locality 48 has a single Curie temperature at 158°C. Because the thermomagnetic curve for this sample is nearly reproducible between heating and cooling and shows no evidence of a disproportionation peak, the oxidation parameter for this sample is probably less than about 0.3 or 0.4 [*Grommé and Mankinen*, 1976]. For a nearly stoichiometric titanomagnetite with this Curie temperature, the ulvöspinel content is about 60%. Two samples have two Curie temperatures, one at about 200° and another at slightly greater than 550°C. These are samples that were incompletely oxidized at high temperature, resulting in two separate magnetic phases. The average Curie temperature for the basaltic rocks from Long Valley is 559° ± 15°C. The titanomagnetites in the more silicic rocks in the late basalt group (localities 31 and 39) have lower Ti contents, as would be expected, and Curie temperatures of about 500°C. The oxide in the andesite at locality 18 has the lowest Ti content, is cation-deficient, and yields a type 4 heating curve similar to the moat rhyolites and rim rhyodacites.

The andesite flows exposed in southernmost Inyo Crater (localities 50, 51, and 52) illustrate the increase in oxide cation deficiency due to weathering. The magnetic directions of these flows (Figure 7*a*) show that all three were erupted in rapid succession. The lowest of the flows (Figure 10*a*) contains a nonstoichiometric oxide, but the originally low Curie temperature (250°C) is still very apparent. The middle flow (Figure 10*b*) is clearly more oxidized, but the presence of a lower Curie temperature can still be seen. The uppermost flow is the most highly oxidized, and little evidence of the original Curie temperature remains (Figure 10*c*).

PALEOMAGNETIC POLE AND PALEOSECULAR VARIATION

All reliable data in Table 1 were used to calculate a mean magnetic field direction, a mean paleomagnetic pole, and associated statistical parameters. Those units thought to be related were omitted from the analyses, as were units whose 95% confidence limits were greater than 9°. Also omitted were the units at localities 15, 16, and 18, which are on the structural block thought to be tilted. No units have virtual geomagnetic pole (VGP) latitudes less than 50°N. Because many eruptions occurred at approximately the same time, an attempt was also made to remove the bias introduced by multiple sampling of the same field direction. Samples from all units within each rock type thought to have been erupted at the same time were averaged to give a mean for that instanta-

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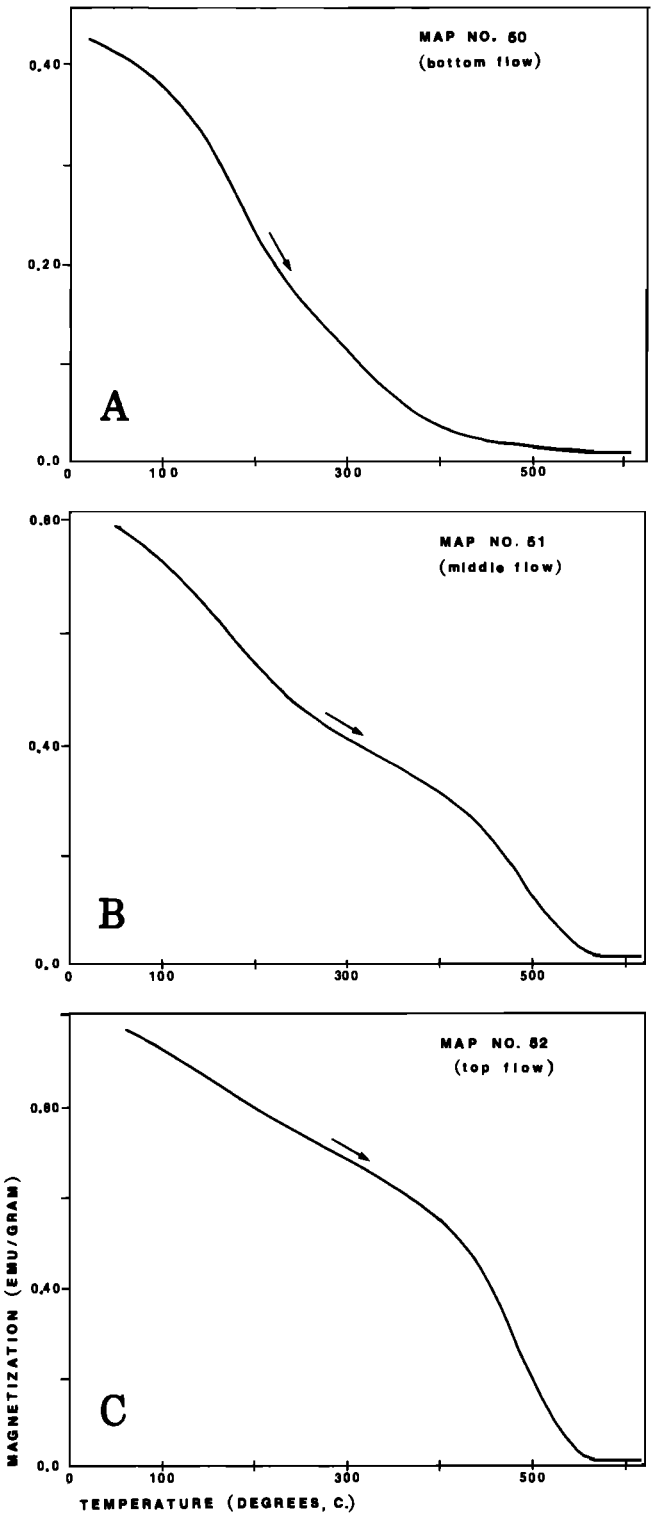


Fig. 10. Strong-field thermomagnetic heating curves for samples from the three andesite flows sampled in South Inyo Crater showing changes presumably due to low-temperature oxidation (weathering).

neous field direction. Results of these averages are given in Table 4; these means were given unit weight in the final calculations. Statistical analyses of all reliable units and all units after averaging for multiple eruptions are given in Table 5.

At many localities it has been found that when a set of VGP's is averaged, the resulting mean pole does not parallel the Earth's rotational axis as it should if, over a long time, the Earth's field is represented by a geocentric axial dipole. *McElhinny and Merrill* [1975] noted that nearly one half of the paleomagnetic poles covering parts of the last 5 m.y. have 95% confidence limits that do not include the geographic pole, although regionally averaged poles nearly always do. Also, many of these discrepant poles are situated on lines of longitude that are approximately 180° away from the mean site longitudes (the "farside" effect). There also seems to be some tendency toward an eastward deflection of declination from the expected value of 0°, although this tendency is not well established. Because the time-averaged geomagnetic field was thought to have zonal symmetry, offsets of the VGP were first thought to be due to a tilt of the dipole field [*Cox and Doell*, 1964] and were later explained as being due to a northward displacement of the axial dipole [*Wilson*, 1970]. *Cox* [1975] proposed an alternative model in which the discrepant poles could be explained by a lack of zonal symmetry in the nondipole field. In this model, the core-surface sources producing the nondipole field are not randomly distributed but occur preferentially at certain latitudes. During periods of normal polarity, flux lines were believed to emerge from low latitudes in the core and reenter at mid-latitudes, producing a field similar to the present-day nondipole field. By rotating the present nondipole field, *Cox* [1975] was able to show a shallowing of magnetic inclinations in low and mid-latitudes (producing farside VGP's) in an amount comparable to that seen in the paleomagnetic data. The difference between the inclination measured at a site and that expected from a geocentric dipole (ΔI) was used by *Cox* [1975] as a measure of the amount of zonal asymmetry. At the latitude of Long Valley, ΔI should be $-2.0°$ according to the *Cox* [1975] model. *Merrill and McElhinney* [1977] show a ΔI of $-3.4°$ ($\sigma = 1.0°$) for normal data in the latitudinal band 30° to 40°N, although in the longitudinal sector between 120° and 300°E, ΔI was found to be $-1.9°$ ($\sigma = 0.6°$). ΔI for Long Valley is $-1.3°$ using all data and $-2.4°$ using the selected data group. While these values are in agreement with the models, neither of the two mean directions is significantly different from the field expected from a geocentric dipole. Similarly, neither of the calculated paleomagnetic pole positions (Table 5 and Figure 11) is distinguishable from the geographic axis. Thus, notwithstanding the periodic and episodic eruptions of these volcanic rocks, the effects of secular variation of the nondipole field have been successfully averaged.

Because virtually all asymmetry of the nondipole field has apparently been averaged out by the Long Valley rocks, the dispersion of these VGP's should give a reasonably accurate measure of the paleosecular variation for this latitude. The dispersion given in Table 5 is the dispersion about the mean directions. While the dispersion of VGP's about the geographic pole is also commonly used, the values for both are nearly identical in this case, differing by only 0.2° for the selected data set. In Table 5, S_T is the total dispersion and S_F is the ancient field dispersion as used by *McElhinney and Merrill* [1975]. S_F has been corrected for the within-site dispersion due to experimental errors, but no correction has been made for the effects of local magnetic anomalies. S_u and S_l are the upper and lower 95% confidence limits, respectively, for S_F calculated using the method of *Cox* [1969].

For all reliable units from Long Valley, the 95% confidence limits about $S_F = 16.0°$ are 19.3° and 13.6°. This is somewhat larger than the value of $S_F = 12.5°$ reported by *Doell and Cox* [1972] for 21 lava flows from the western United States. However, *McElhinny and Merrill* [1975] report an almost identical value of $S_F = 16.3°$ ($S_u = 18.4°$, $S_l = 14.7°$) for the western

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TABLE 5. Statistical Analysis of Flow Average Data

Data Analyzed	Latitude	Longitude	N	R	K	α_{95}	S_T	S_F	S_u	S_l
All units	89.7	138.4	33	31.7280	25	5.1	16.3	16.0	19.3	13.6
Selected units	87.6	006.1	23	22.1476	26	6.1	16.1	15.9	20.1	13.2
Data Analyzed	Inclination	Declination	N	R	K	α_{95}	S_T	S_F	S_u	S_l
All units	55.8	359.4	33	32.0834	35	4.3	13.8	13.6	16.4	11.6
Selected units	54.7	002.5	23	22.3779	35	5.2	13.7	13.6	17.2	11.3

Latitude and Longitude, north latitude and east longitude, respectively, of paleomagnetic pole; N, number of units analyzed; R, resultant vector; K, precision parameter [*Fisher*, 1953]; α_{95}, 95% confidence cone about mean pole; S_T, total angular dispersion about mean pole; S_F, ancient field dispersion (experimental errors removed); S_u and S_l, upper and lower limits, respectively, of S_F from *Cox* [1969].

United States on the basis of 76 units spanning the last 5 m.y. *Champion* [1980] reports a total dispersion S_T of 12.2° for 77 Holocene lave flows from the western United States. The very low dispersion reported by *Champion* [1980] combined with the limited Recent data summarized by *McElhinny and Merrill* [1975] indicate that very recently the geomagnetic field has been unusually quiet. The fact that Brunhes age (<0.7 m.y.) volcanic rocks from Long Valley show the same dispersion as that reported for the same region for the last 5 m.y. by *McElhinny and Merrill* [1975] tends to support this idea. The 21 lava flows reported by *Doell and Cox* [1972] may not have adequately sampled the total range of dispersion for this region.

Discussion

No evidence was found for the Blake reversed episode, which was proposed on the basis of reversed magnetic inclinations in sedimentary cores from the western North Atlantic Ocean [*Smith and Foster*, 1969; *Denham*, 1976; *Denham et al.*, 1977]. The reversed inclinations found in these cores occur within the late Pleistocene X faunal zone of *Ericson et al.* [1961] and represent a portion of the interval between 0.075 and 0.125 m.y. ago [*Denham*, 1976]. A record of what may also be the Blake episode has been found in an unoriented core that penetrated a sequence of offshore clay near Reggio de Calabria in southern Italy [*Creer et al.*, 1980]. More recently, *Nishimura et al.* [1984] have described a reversely magnetized tuff in the Sigura-gura Formation near Lake Toba, Sumatra, that has yielded ages of 75,000 years (K-Ar) and 0.10 ± 0.2 m.y. (fission track), and which they correlate with the Blake episode. Paleomagnetic measurements performed on samples from the Lake Toba tuff are reported in detail by *Sasajima et al.* [1984], who also review these and other currently available evidence for the Blake episode. The fact that no reversed volcanic units of this age were found in Long Valley may be interpreted in several ways. First, the Blake episode might have been a regional rather than global phenomenon [*Denham*, 1976], although the results from Italy and Indonesia, if confirmed, suggest the possibility of a global event at about 0.1 m.y. *Sasajima et al.* [1984] consider that the balance of evidence does favor a global geomagnetic event. The numerous volcanic eruptions in the west moat of Long Valley caldera from about 0.115 m.y. to about 0.095 m.y. ago, all of which record normal polarity, may therefore indicate that the Blake episode occurred just prior to or after this time interval. Second, it is possible that the episodic nature of the volcanism is a factor because we have shown that the eruptive periods in the west moat may have been separated by approximately 10,000 years. Some short events may have a duration of 10,000 years or less [*Mankinen and Grommé*, 1982], and the Blake episode itself has been estimated to have lasted about 5,000 years [*Smith and Foster*, 1969; *Denham*, 1976]. Finally, it is possible that the Blake episode was recorded by one of the numerous Long Valley eruptions but that the products of that eruption were not sampled during this study. Lacustrine silts associated with a tephra layer in Long Valley have reversed polarity [*Liddicoat et al.*, 1980]. This reversal definitely occurred during the Brunhes chron, but its absolute age is not known. However, trace-element chemistry of the tephra shows a close affinity to the 0.05- to 0.150-m.y. old quartz latites of Mammoth Mountain [*Liddicoat et al.*, 1980].

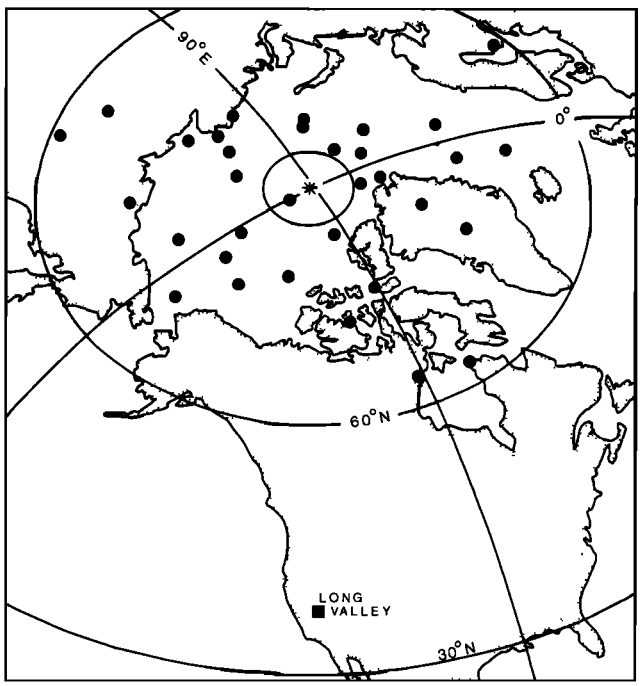


Fig. 11. VGP positions (dots) of reliable volcanic units from Long Valley caldera. Asterisk, overall mean with its associated 95% confidence limit. Portion of equal-area projection.

Potassium-argon dating has demonstrated a periodicity in eruptive cycles in large silicic volcanic centers such as the Valles caldera, New Mexico [*Doell et al.*, 1968], the Geysers–Clear Lake area, California [*Donnelly*, 1977; *Donnelly-Nolan et al.*, 1981] and Long Valley [*Bailey et al.*, 1976]. However, the resolution of the K-Ar method is often unable to determine the pattern of eruption within each period of activity.

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Because secular changes in the geomagnetic field occur fairly rapidly, paleomagnetic studies can distinguish much shorter intervals of time (10^2 to 10^3 years) than can radiometric dating methods. The present study and studies in the Clear Lake area [*Mankinen et al.*, 1981] show that many eruptions can occur during very short periods of time. Inspection of the paleomagnetic data for the ring domes of the Valles caldera [*Doell et al.*, 1968], which are analogous to the moat rhyolites of Long Valley, suggests that the pattern of eruption may be quite similar in both areas. The oldest ring domes sampled in the Valles caldera (sites S34 and S35) are statistically different in age and in paleomagnetic direction. With the next younger group, sites S37 and S38 give identical ages, but the magnetic directions are distinctly different (normal and intermediate polarity). However, these units were erupted during the transition from normal to reversed polarity at the end of the Jaramillo Normal Subchron [*Doell and Dalrymple*, 1966; *Doell et al.*, 1968], and the time involved may be no more than a few tens or hundreds of years [*Mankinen et al.*, 1981]. Within the group of Valles ring domes that is approximately 0.7 m.y. old, sites S40 and S41 show little difference in age and have identical directions of magnetization, which suggests that these domes may have erupted at about the same time. Their eruption was preceded by the eruption at site S39 approximately 20,000 years earlier. Combining the dates and paleomagnetic directions of the approximately 0.5-m.y.-old group of domes suggests that the eruption at site S42 was followed some 40,000 years later by those at sites S34 and S44, which in turn were followed by the dome at site S45 approximately 30,000 years later.

For these two studies the eruptive pattern suggests that within each long (10^5 year) period of active volcanism, shorter episodes of eruption also occur that are separated by a few tens of thousands of years. These shorter episodes may be single eruptions, or they may consist of a few eruptions occurring almost simultaneously, separated by no more than a few hundred years. Detailed K-Ar and paleomagnetic studies in other large silicic centers will be very useful in determining if the pattern suggested here is a feature common to other such areas.

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